David Polefrone

Curriculum Vitae

Education

2021–pres. **PhD (in progress)**, *University of Pennsylvania*, Philadelphia, *GPA – 3.77*.

2017–2021 **Bachelor of Science in Chemistry**, *University of Maryland*, College Park, *GPA – 3.95*.

2017–2021 **Bachelor of Science in Economics**, *University of Maryland*, College Park, Dual Degree.

Team Honors Thesis

Title *Investigating the Application of Interpretability Techniques to Computational Toxicology*

Supervisors Asst. Professor Soheil Feizi & Professor Aravind Srinivasan

Description Worked with a team of undergraduate students in the Gemstone Honors College at UMD to apply interpretability techniques to toxicological machine learning models. Developed a cheminformatics pipeline for generating descriptors for small molecules in the Tox21 dataset.

Experience

Research

2021–Present **1st Year Graduate Student**, Kozlowski Group, Penn.
Exploring machine learning (ML) techniques to predict regioselectivity for Minisci-type reactions (collaborative effort with GSK medicinal chemistry). Working to develop ML-based quantitative structure-selectivity relationship methods for dimeric zinc catalysts.

2019–2021 **Undergraduate Research Assistant**, Gutierrez Group, UMD.
Used computational tools to predict, understand, and elucidate organic reaction mechanisms. Employed density functional theory (DFT) methods to simulate nickel-catalyzed cross-coupling reactions.

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2018–2019 **Undergraduate Research Assistant**, Rodriguez Group, UMD.
Created data analysis scripts in Python for data collected from analytical chemistry techniques.

Internships and Work Experience

Summer 2021 **Contract Analyst**, DeciBio Consulting, Remote.
Conducted market research for a multinational biotechnology client interested in expanding NGS capabilities.

Fall 2020 **Student Consultant**, QUEST Honors Program, UMD.
Provided process design and sustainability solutions as part of a five-person team to a Fortune 500 company in the chemical industry. Won "Most Outstanding Capstone" award based on our recommendations.

Summer 2020 **Summer Analyst**, Point72 Asset Management, Remote.
Developed financial modelling and pitch skills for discretionary long/short equity trading.

Summer 2019 **Medical Affairs Intern**, Roche Tissue Diagnostics, Tucson, AZ.
Investigated the cost-effectiveness of various diagnostic technologies to screen for hereditary colorectal cancer.

Summer 2017 **Research Intern**, NextLOGiK, Inc., Columbia, MD.
Conducted research on federal and state regulations pertaining to agricultural business.

Extracurriculars

2018–2021 **Chief of Staff**, *UMD Student Government Association*.
Provided direct support and advice to the student body president on strategic issues. Coordinated initiatives for 11 committee directors across an organization of roughly 150 students. Previously served as an academic legislator and parliamentarian.

2018–2021 **Events and Lectures Chair**, *Omicron Delta Kappa Leadership Honors Society*.
Organized annual speaker series with a dedicated budget for attracting acclaimed lecturers. Previously served as the Public Relations Chair.

2018–2019 **VP of Finance**, *Maryland Parliamentary Debate Team*.
Maintained the team's finances for tournament registration and submitted funding requests. Competed in tournaments across the Northeast.

Awards

2021 Maryland Medallion Society (awarded to top 20 seniors), UMD

2021 Senior Marshal, UMD

2017–2021 President's Scholarship, UMD

2021 Canadian Royal Society of Chemistry Certificate of Excellence, UMD Department of Chemistry

2020 Bruce B. Jarvis Outstanding Junior Award, UMD Department of Chemistry

2018 Top Ten Freshman, Omicron Delta Kappa Honors Society at UMD

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2018 Rookie of the Year, UMD Gemstone Honors College

Computer skills

Basic Java, HTML, Adobe InDesign

Intermediate LaTeX, Linux, Matlab, Stata, R

Advanced Python, Microsoft Windows Suite

Scientific RDKit, OpenBabel, NumPy, SciPy, TensorFlow, Gaussian 09/16, ORCA, SCINE Sparrow, ChemDraw

Communication Skills

2017 National Finalist, Extemporaneous Speaking, NCFL.

2017 Delegate to the U.S. Senate Youth Program.

2016 Delegate to Boys Nation.

Languages

English **Native**

French **Intermediate** *Reading, writing*

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